

Full Application ERD

Source: docs/database/application-erd.md

```
# Full Application ERD

```mermaid
erDiagram
 ENTERPRISE {
 string id PK
 string enterprise_name
 }
 APP_USER {
 int id PK
 string full_name
 string email
 string role_name
 string status
 }
 COMPANY {
 int id PK
 string company_code
 string company_name
 }
 LOCATION {
 int id PK
 string location_code
 string location_name
 string state
 }
 FUND {
 int id PK
 string fund_code
 string fund_name
 }
 FUNCTION_MASTER {
 int id PK
 string function_code
 string function_name
 }
 PROJECT {
 int id PK
 string project_code
 string project_name
 }
 PROJECT_LOCATION_MAP {
 int id PK
```

```
 int project_id FK
 int location_id FK
}
FINANCIAL_YEAR {
 int id PK
 string fy_code
 date start_date
 date end_date
}
GRANT_MASTER {
 string id PK
 string grant_code
 string grant_name
 string grantor_name
}
GRANT_MOU {
 string id PK
 string grant_id FK
}
GRANT_BUDGET {
 string id PK
 string grant_id FK
 int project_id FK
 int location_id FK
}
GRANT_FUC {
 string id PK
 string grant_id FK
 int financial_year_id FK
}
GRANT_REPORT {
 string id PK
 string grant_id FK
 int project_id FK
}
GRANT_COA_MAP {
 string id PK
 string grant_id FK
 int ledger_id FK
}
GRANT_UTILIZATION {
 string id PK
 string grant_id FK
 int location_id FK
 int project_id FK
}
GRANT_ALLOCATION {
 string id PK
 string grant_id FK
```

```
 int location_id FK
 date period_start
 date period_end
}
TDS_RULE {
 int id PK
 int financial_year_id FK
 string section_code
 decimal tds_rate
}
COA_MASTER {
 int id PK
 string master_name
}
COA_GROUP {
 int id PK
 int coa_master_id FK
 string group_name
}
COA_LEDGER {
 int id PK
 int coa_group_id FK
 string ledger_name
 boolean is_party_account
}
EXPENSE_MAPPING {
 int id PK
 string expense_type
 int fund_id FK
 string grant_id FK
 int function_id FK
 int project_id FK
 int ledger_id FK
 int tds_rule_id FK
}

CONSULTANT_REQUISITION {
 string id PK
 string vendor_name
 int location_id FK
 int project_id FK
 int fund_id FK
 int tds_rule_id FK
}
CONSULTANT_TDS_ACCOUNT {
 string id PK
 string consultant_requisition_id FK
 int tds_rule_id FK
}
```

```
CONSULTANT_VOUCHER {
 string id PK
 string consultant_requisition_id FK
}
CONSULTANT_JV_MAP {
 string id PK
 string consultant_requisition_id FK
 int ledger_id FK
}

TRAVEL_REQUISITION {
 string id PK
 int user_id FK
 int location_id FK
 int project_id FK
 int fund_id FK
 string grant_id FK
 int function_id FK
}
TRAVEL_EXPENSE {
 string id PK
 string travel_requisition_id FK
 decimal actual_ticket_cost
 decimal actual_lodging_cost
 decimal actual_local_conveyance
}
TRAVEL_PER_DIEM_POLICY {
 string id PK
 int financial_year_id FK
 int location_id FK
}
TRAVEL_VOUCHER {
 string id PK
 string travel_requisition_id FK
}
TRAVEL_JV_MAP {
 string id PK
 string travel_requisition_id FK
 int ledger_id FK
}

RENT_REQUISITION {
 string id PK
 int location_id FK
 int fund_id FK
 int tds_rule_id FK
 string landlord_name
}
RENT_TDS_ACCOUNT {
```

```
 string id PK
 string rent_requisition_id FK
 int tds_rule_id FK
}
RENT_VOUCHER {
 string id PK
 string rent_requisition_id FK
}
RENT_JV_MAP {
 string id PK
 string rent_requisition_id FK
 int ledger_id FK
}
REQUISITION_APPROVAL {
 string id PK
 string source_type
 string source_id
 int approver_user_id FK
 string status
}

EXPENSE_VOUCHER {
 string id PK
 int location_id FK
 int fund_id FK
 int ledger_id FK
 string source_type
 string source_id
}
BANK_VOUCHER {
 string id PK
 int location_id FK
 int fund_id FK
 string payee_name
 string source_type
 string source_id
}
VOUCHER_APPROVAL {
 string id PK
 string voucher_type
 string voucher_id
 int approver_user_id FK
 string status
}
BANK_RECON_UPLOAD {
 string id PK
 int location_id FK
 date statement_date
}
```

```
BANK_RECON_MAP {
 string id PK
 string upload_id FK
 string bank_voucher_id FK
 string expense_voucher_id FK
}
CASHFLOW_TRANSACTION {
 string id PK
 int location_id FK
 int financial_year_id FK
 string source_type
 string source_id
}
BUDGET_ENTRY {
 string id PK
 int financial_year_id FK
 int location_id FK
 int fund_id FK
 int project_id FK
}
FORECAST_ENTRY {
 string id PK
 string budget_entry_id FK
 date forecast_month
}

FIXED_ASSET_PURCHASE {
 string id PK
 string asset_id
 int location_id FK
 int fund_id FK
 string vendor_name
}
FIXED_ASSET_REGISTER {
 string asset_id PK
 string purchase_id FK
 int location_id FK
 int fund_id FK
 string asset_type
}
FIXED_ASSET_DISPOSAL {
 string id PK
 string asset_id FK
 date disposal_date
}
DEPRECIATION_RATE_IT {
 string id PK
 int financial_year_id FK
 string asset_type
}
```

```
}
DEPRECIATION_RATE_CA {
 string id PK
 int financial_year_id FK
 string asset_type
}
DEPRECIATION_ENTRY {
 string id PK
 string asset_id FK
 int financial_year_id FK
}

BENEFICIARY_BATCH {
 string id PK
 int location_id FK
 string batch_number
}
BENEFICIARY {
 string id PK
 int location_id FK
 string batch_id FK
 string beneficiary_name
}
BENEFICIARY_FEE {
 string id PK
 int location_id FK
 decimal fee_amount
}
BENEFICIARY_COLLECTION {
 string id PK
 string beneficiary_id FK
 decimal amount_collected
 date collection_date
}
BENEFICIARY_FUNDER_ALLOCATION {
 string id PK
 string batch_id FK
 string grant_id FK
}

EMPLOYEE {
 int id PK
 int location_id FK
 string employee_name
 string department
}
PAYROLL_RUN {
 string id PK
 int financial_year_id FK
```

```
 string payroll_month
 }
 PAYROLL_ENTRY {
 string id PK
 string payroll_run_id FK
 int employee_id FK
 int fund_id FK
 decimal gross_salary
 decimal net_salary
 }

 PROCUREMENT_VENDOR {
 int id PK
 string vendor_name
 string category
 string status
 }

 PROCUREMENT_REQUISITION {
 int id PK
 int fund_id FK
 int project_id FK
 int requested_by_user_id FK
 decimal amount
 }

 PROCUREMENT_QUOTATION {
 int id PK
 int requisition_id FK
 int vendor_id FK
 decimal quote_amount
 }

 PURCHASE_ORDER {
 int id PK
 int requisition_id FK
 int vendor_id FK
 int quotation_id FK
 int fund_id FK
 }

 GOODS_RECEIPT_NOTE {
 int id PK
 int purchase_order_id FK
 date received_on
 string match_status
 }

 AUDIT_LOG {
 string id PK
 string table_name
 string record_id
 int action_by FK
 }
```



```

 datetime action_at
}

ENTERPRISE ||--o{ APP_USER : has
ENTERPRISE ||--o{ COMPANY : owns
ENTERPRISE ||--o{ LOCATION : owns
ENTERPRISE ||--o{ FUND : owns
ENTERPRISE ||--o{ FUNCTION_MASTER : owns
ENTERPRISE ||--o{ PROJECT : owns
ENTERPRISE ||--o{ FINANCIAL_YEAR : owns
ENTERPRISE ||--o{ GRANT_MASTER : owns
ENTERPRISE ||--o{ TDS_RULE : owns
ENTERPRISE ||--o{ COA_MASTER : owns
ENTERPRISE ||--o{ EXPENSE_MAPPING : owns
ENTERPRISE ||--o{ AUDIT_LOG : logs

PROJECT ||--o{ PROJECT_LOCATION_MAP : spans
LOCATION ||--o{ PROJECT_LOCATION_MAP : assigned_to

COA_MASTER ||--o{ COA_GROUP : contains
COA_GROUP ||--o{ COA_LEDGER : contains

FINANCIAL_YEAR ||--o{ TDS_RULE : governs
FUND ||--o{ EXPENSE_MAPPING : filters
GRANT_MASTER ||--o{ EXPENSE_MAPPING : filters
FUNCTION_MASTER ||--o{ EXPENSE_MAPPING : filters
PROJECT ||--o{ EXPENSE_MAPPING : filters
COA_LEDGER ||--o{ EXPENSE_MAPPING : posts_to
TDS_RULE ||--o{ EXPENSE_MAPPING : taxes

GRANT_MASTER ||--o{ GRANT_MOU : has
GRANT_MASTER ||--o{ GRANT_BUDGET : has
GRANT_MASTER ||--o{ GRANT_FUC : has
GRANT_MASTER ||--o{ GRANT_REPORT : has
GRANT_MASTER ||--o{ GRANT_COA_MAP : maps
GRANT_MASTER ||--o{ GRANT_UTILIZATION : tracks
GRANT_MASTER ||--o{ GRANT_ALLOCATION : allocates
PROJECT ||--o{ GRANT_BUDGET : budgets
LOCATION ||--o{ GRANT_BUDGET : budgets
FINANCIAL_YEAR ||--o{ GRANT_FUC : reports
PROJECT ||--o{ GRANT_REPORT : reports
COA_LEDGER ||--o{ GRANT_COA_MAP : maps
LOCATION ||--o{ GRANT_UTILIZATION : utilized_at
PROJECT ||--o{ GRANT_UTILIZATION : utilized_for
LOCATION ||--o{ GRANT_ALLOCATION : allocated_to

LOCATION ||--o{ CONSULTANT_REQUISITION : requested_at
PROJECT ||--o{ CONSULTANT_REQUISITION : charged_to
FUND ||--o{ CONSULTANT_REQUISITION : funded_by

```

```

TDS_RULE ||--o{ CONSULTANT_REQUISITION : taxed_by
CONSULTANT_REQUISITION ||--o{ CONSULTANT_TDS_ACCOUNT : has
CONSULTANT_REQUISITION ||--o{ CONSULTANT_VOUCHER : creates
CONSULTANT_REQUISITION ||--o{ CONSULTANT_JV_MAP : maps
TDS_RULE ||--o{ CONSULTANT_TDS_ACCOUNT : applies
COA_LEDGER ||--o{ CONSULTANT_JV_MAP : posts_to

```

```

APP_USER ||--o{ TRAVEL_REQUISITION : requests
LOCATION ||--o{ TRAVEL_REQUISITION : occurs_at
PROJECT ||--o{ TRAVEL_REQUISITION : charged_to
FUND ||--o{ TRAVEL_REQUISITION : funded_by
GRANT_MASTER ||--o{ TRAVEL_REQUISITION : grant_backed
FUNCTION_MASTER ||--o{ TRAVEL_REQUISITION : functional_area
TRAVEL_REQUISITION ||--o{ TRAVEL_EXPENSE : settles
TRAVEL_REQUISITION ||--o{ TRAVEL_VOUCHER : creates
TRAVEL_REQUISITION ||--o{ TRAVEL_JV_MAP : maps
FINANCIAL_YEAR ||--o{ TRAVEL_PER_DIEM_POLICY : defines
LOCATION ||--o{ TRAVEL_PER_DIEM_POLICY : applies_at
COA_LEDGER ||--o{ TRAVEL_JV_MAP : posts_to

```

```

LOCATION ||--o{ RENT_REQUISITION : occurs_at
FUND ||--o{ RENT_REQUISITION : funded_by
TDS_RULE ||--o{ RENT_REQUISITION : taxed_by
RENT_REQUISITION ||--o{ RENT_TDS_ACCOUNT : has
RENT_REQUISITION ||--o{ RENT_VOUCHER : creates
RENT_REQUISITION ||--o{ RENT_JV_MAP : maps
TDS_RULE ||--o{ RENT_TDS_ACCOUNT : applies
COA_LEDGER ||--o{ RENT_JV_MAP : posts_to

```

```

APP_USER ||--o{ REQUISITION_APPROVAL : approves

```

```

LOCATION ||--o{ EXPENSE_VOUCHER : booked_at
FUND ||--o{ EXPENSE_VOUCHER : booked_to
COA_LEDGER ||--o{ EXPENSE_VOUCHER : ledger
LOCATION ||--o{ BANK_VOUCHER : paid_from
FUND ||--o{ BANK_VOUCHER : paid_from
APP_USER ||--o{ VOUCHER_APPROVAL : approves
EXPENSE_VOUCHER ||--o{ VOUCHER_APPROVAL : approval_flow
BANK_VOUCHER ||--o{ VOUCHER_APPROVAL : approval_flow
BANK_RECON_UPLOAD ||--o{ BANK_RECON_MAP : maps
BANK_VOUCHER ||--o{ BANK_RECON_MAP : matches
EXPENSE_VOUCHER ||--o{ BANK_RECON_MAP : matches

```

```

LOCATION ||--o{ CASHFLOW_TRANSACTION : tracks
FINANCIAL_YEAR ||--o{ CASHFLOW_TRANSACTION : tracks
FINANCIAL_YEAR ||--o{ BUDGET_ENTRY : budgets
LOCATION ||--o{ BUDGET_ENTRY : budgets
FUND ||--o{ BUDGET_ENTRY : budgets
PROJECT ||--o{ BUDGET_ENTRY : budgets

```

```

BUDGET_ENTRY ||--o{ FORECAST_ENTRY : forecasts

LOCATION ||--o{ FIXED_ASSET_PURCHASE : bought_at
FUND ||--o{ FIXED_ASSET_PURCHASE : funded_by
FIXED_ASSET_PURCHASE ||--|| FIXED_ASSET_REGISTER : creates
LOCATION ||--o{ FIXED_ASSET_REGISTER : held_at
FUND ||--o{ FIXED_ASSET_REGISTER : funded_by
FIXED_ASSET_REGISTER ||--o{ FIXED_ASSET DISPOSAL : disposed_by
FINANCIAL_YEAR ||--o{ DEPRECIATION_RATE_IT : defines
FINANCIAL_YEAR ||--o{ DEPRECIATION_RATE_CA : defines
FIXED_ASSET_REGISTER ||--o{ DEPRECIATION_ENTRY : depreciates
FINANCIAL_YEAR ||--o{ DEPRECIATION_ENTRY : posts

LOCATION ||--o{ BENEFICIARY_BATCH : runs
LOCATION ||--o{ BENEFICIARY : enrolls
BENEFICIARY_BATCH ||--o{ BENEFICIARY : groups
LOCATION ||--o{ BENEFICIARY_FEE : fee_policy
BENEFICIARY ||--o{ BENEFICIARY_COLLECTION : pays
BENEFICIARY_BATCH ||--o{ BENEFICIARY_FUNDER_ALLOCATION : funded_by
GRANT_MASTER ||--o{ BENEFICIARY_FUNDER_ALLOCATION : funds

LOCATION ||--o{ EMPLOYEE : employs
FINANCIAL_YEAR ||--o{ PAYROLL_RUN : runs
PAYROLL_RUN ||--o{ PAYROLL_ENTRY : contains
EMPLOYEE ||--o{ PAYROLL_ENTRY : paid_to
FUND ||--o{ PAYROLL_ENTRY : charged_to

FUND ||--o{ PROCUREMENT_REQUISITION : budgets
PROJECT ||--o{ PROCUREMENT_REQUISITION : for_project
APP_USER ||--o{ PROCUREMENT_REQUISITION : requested_by
PROCUREMENT_REQUISITION ||--o{ PROCUREMENT_QUOTATION : invites
PROCUREMENT_VENDOR ||--o{ PROCUREMENT_QUOTATION : quotes
PROCUREMENT_REQUISITION ||--o{ PURCHASE_ORDER : converts_to
PROCUREMENT_VENDOR ||--o{ PURCHASE_ORDER : receives
PROCUREMENT_QUOTATION ||--o{ PURCHASE_ORDER : selected_as
FUND ||--o{ PURCHASE_ORDER : charged_to
PURCHASE_ORDER ||--o{ GOODS_RECEIPT_NOTE : received_as

APP_USER ||--o{ AUDIT_LOG : acts
...

```